Lab 4

Switch and Break Statements in C++

Learning Objectives

After completing this lab, after completing this lab we learnt:

- To understand the working of the switch statement in C++.
- To learn how to use the break statement inside switch cases.
- To implement decision-making programs using switch-case.
- To replace multiple if-else conditions using switch structure.

Switch Statement

The switch statement is a control structure in C++ used for decision making. It allows the execution of one block of code among many alternatives based on the value of a variable.

The break statement is used to terminate a case inside the switch. Without break, the program continues execution into the next cases (called fall-through).

Syntax:

```
switch(expression) { //Condition should be true , then cases runs
    case value1:
        // code block
        break;
    case value2:
        // code block
        break;
    default:
        // code block
}
```

Explanation:

• switch(expression) → checks variable value
• case → defines different conditions
• break → stops execution of further cases
• default → executes when no case matches

Example: Program Example (Switch Case with Break). Simple Calculator using Switch

```
int main() {
    int a, b, choice;         // Variables for numbers and user choice
    cout << "Enter first number: ";     // Input first number
    cin >> a;
    cout << "Enter second number: ";  // Input second number
    cin >> b;
    cout << "Select operation: ";       // Show menu
    cout << "\n1. Addition";
    cout << "\n2. Subtraction";
    cout << "\n3. Multiplication";
    cout << "\nEnter choice: ";
    cin >> choice;                      // Read user choice
    switch(choice) {                    // Start switch statement
        case 1:                         // Addition case
            cout << "Sum = " << a + b << endl;  // Add numbers
            break;                      // Exit switch
        case 2:                         // Subtraction case
            cout << "Difference = " << a - b << endl;
            break;                      // Exit switch
        case 3:                         // Multiplication case
            cout << "Product = " << a * b << endl;
            break;                      // Exit switch
        default:                        // Invalid input case
            cout << "Invalid choice!" ;  return 0;    // End program
```

Input:

Enter first number: 10

Enter second number: 5

Select operation: 1

Output:

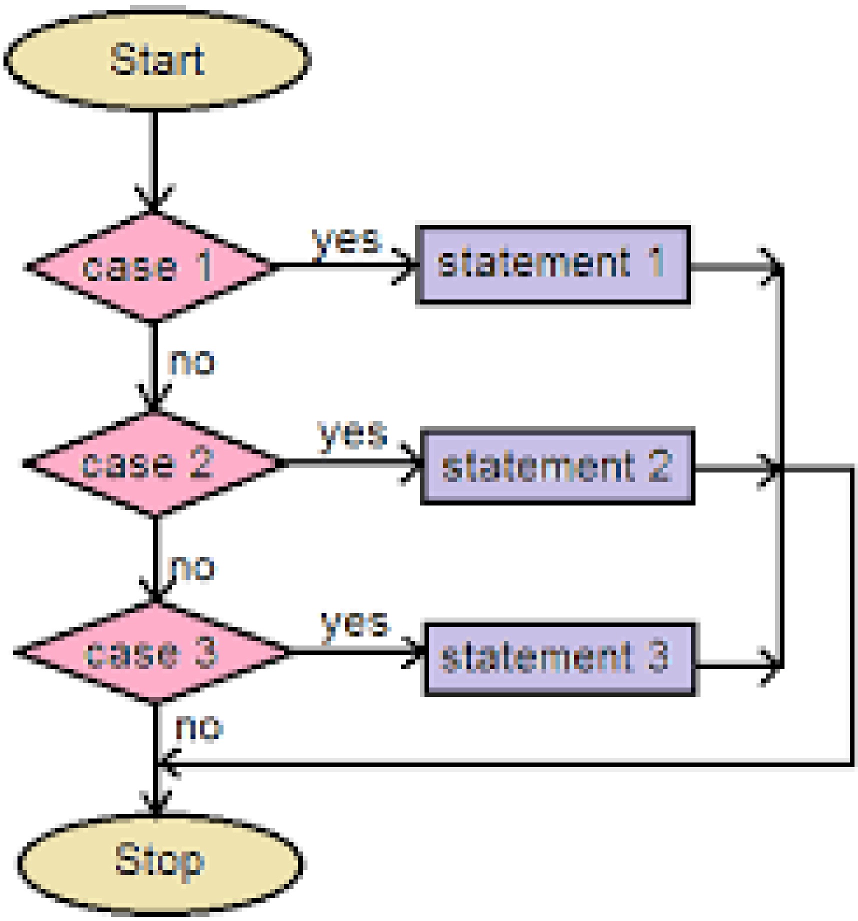
Sum = 15

Example: Day of Week using Switch

```
#include <iostream>      // Input/output library
using namespace std;     // Standard namespace
int main() {
    int day;                      // Variable for day number
    cout << "Enter day number (1-7): ";
    cin >> day;
    switch(day) {          // Switch on day value
        case 1:
            cout << "Monday" << endl;
            break;
        case 2:
            cout << "Tuesday" << endl;
            break;
        case 3:
            cout << "Wednesday" << endl;
            break;
        case 4:
```

```
            cout << "Thursday" << endl;
            break;
        case 5:
            cout << "Friday" << endl;
            break;
        case 6:
            cout << "Saturday" << endl;
            break;
        case 7:
            cout << "Sunday" << endl;
            break;
        default:
            cout << "Invalid day number!" << endl;
    }
  return 0;
}
```

Flow Chart:



Continue Statement in C++

The continue statement is used inside loops to skip the current iteration and move directly to the next iteration of the loop.

It does not terminate the loop like break; instead, it only skips one step and continues execution.

Syntax:

```
continue;
```

When continue is executed inside a loop:

- Remaining code in the current iteration is skipped
- Control jumps to the next loop iteration

Examples:

```
#include <iostream>                    // Input/output library
using namespace std;                   // Standard namespace
int main() {
    for (int i = 1; i <= 5; i++) {     // Loop from 1 to 5
        if (i == 3) {                  // If i is 3
            continue;                  // Skip this iteration
        }
        cout << i << endl;             // Print value
    }
    return 0;                          // End program
}
```

Output:

1

2

3

4

Comparison of Break and Continues statement:

Break	Continue
Break is used to **terminate** the execution of the loop.	Continue is not used to terminate the execution of loop.
It **breaks** the iteration.	It **skips** the iteration.
When this statement is executed, control will come out from the loop and executes the statement immediate after loop.	When this statement is executed, it will not come out of the loop but moves/jumps to the next iteration of loop.
Break is used with loops as well as switch case.	Continue is only used in loops, it is not used in switch case.

Lab5: Lab Tasks

Task 1: Write a C++ program that takes an input of a day's number and uses a switch statement to output the corresponding day's name.

```
#include <iostream>              // Header file
using namespace std;             // To use cin & cout without std::
int main()   {                // Main function starts
    int day;                              // Variable to store day number
    cout << "Enter day number (1-7): ";   // Ask user for input
    cin >> day;                                // Take input from user
    switch(day) {     // Switch checks entered day number
        case 1:                                   // If user enters 1
            cout << "Monday";                // Display Monday
```

```
 break;                              // Exit switch
   case 2:                           // If user enters 2
       cout << "Tuesday";            // Display Tuesday
       break;                        // Exit switch
   case 3:                           // If user enters 3
       cout << "Wednesday";          // Display Wednesday
       break;                        // Exit switch
   case 4:                           // If user enters 4
       cout << "Thursday";           // Display Thursday
       break;                        // Exit switch
   case 5:                           // If user enters 5
       cout << "Friday";             // Display Friday
       break;                        // Exit switch
   case 6:                           // If user enters 6
       cout << "Saturday";           // Display Saturday
       break;
   case 7:                      // If user enters 7
       cout << "Sunday";        // Display Sunday
       break;                   // Exit switch
   default:                     // If user enters wrong number
       cout << "Invalid day number";}      // Show error
return 0;                                  // Program ends
```

Task 2: **Write C++ program using switch that takes a character grade (A,B,C,D) from the user and prints A-Exelent, B-Very Good, C-Good, D-Pass**

```
#include <iostream>    // Header file
using namespace std;   // Standard namespace
int main() {                                  // Main function
    char grade;                               // Variable to store grade
    cout << "Enter your grade (A/B/C/D/F): ";   // Ask user
    cin >> grade;                             // Take grade input
    switch(grade){                            // Check entered grade
        case 'A':                             // If grade is A
            cout << "Excellent";              // Output message
            break;                            // Exit switch
        case 'B':                             // If grade is B
            cout << "Very Good";              // Output message
            break;                            // Exit switch
        case 'C':                             // If grade is C
            cout << "Good";                   // Output message
            break;                            // Exit switch
        case 'F':                             // If grade is F
            cout << "Fail";                   // Output message
            break;                            // Exit switc
        default:       /                      / Invalid input
            cout << "Invalid Grade";}   // Show error
    return 0; }                               // End program
```

Task 3: **Write a C++ program using switch that displays the following menu**

1-convert km to m, 2-convert m to cm, 3-kg to g, 4-celsius to Fahrenheit

```
#include <iostream>    // Header file

int main() {            // Main function

    int choice;        // Variable for menu choice

    float value;       // Variable for entered value

    float result;      // Variable for converted result

    cout << "Conversion Menu" << endl;      // Display heading

    cout << "1. Kilometers to Meters" << endl;   // Option 1

    cout << "2. Meters to Centimeters" << endl;  // Option 2

    cout << "3. Kilograms to Grams" << endl;     // Option 3

    cout << "4. Celsius to Fahrenheit" << endl;  // Option 4

    cout << "Enter your choice: ";       // Ask user for choice

    cin >> choice;                              // Take choice input

    cout << "Enter value: ";               // Ask for value

    cin >> value;                               // Take value input

    switch(choice) {                            // Check selected option

        case 1:                                 // Kilometers to Meters

            result = value * 1000;           // Conversion formula

            cout << "Result = " << result << " meters";   // Display
```

```
break;                                   // Exit
case 2:                                  // Meters to Centimeters
        result = value * 100;            // Conversion formula
        cout << "Result = " << result << " centimeters"; // Display
        break;                           // Exit switch
    case 3:                              // Kilograms to Grams
        result = value * 1000;           // Conversion formula
        cout << "Result = " << result << " grams";    // Display
        break;                           // Exit switch
    case 4:                              // Celsius to Fahrenheit
        result = (value * 9 / 5) + 32;   // Conversion formula
        cout << "Result = " << result << " Fahrenheit"; // Display
        break;                           // Exit switch
    default:                             // Invalid menu option
        cout << "Invalid choice";        // Show error
  }
  return 0;    }                         // Program ends
```